

XANDER DUKAS

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Seeking a Mechanical Engineering Co-op — available Summer–Fall 2027 (July–December)

EDUCATION

Northeastern University

B.S. in Mechanical Engineering

GPA: 3.9 / 4.0 · Dean's List (every semester)

Boston, MA

Expected May 2029

Relevant Coursework: Thermodynamics, Statics & Mechanics of Materials, Differential Equations, Data Structures, Object-Oriented Programming

Norfolk Academy

Head of the Honor Council · Head of the Robotics Team

Norfolk, VA

EXPERIENCE

Engineering Intern — Controls Corporation of America

Virginia Beach, VA — gas pressure regulators (pressure & flow systems)

Jun–Aug 2026

PROJECTS

RxManager — IoT Medication Dispenser

github.com/Geordorini/rxmanager

- Designed and built an automated pill-dispensing system spanning ESP32/MicroPython firmware, a PostgreSQL (Supabase) backend, and a Vercel-deployed dashboard.
- Wrote the device firmware: NTP clock sync, REST polling for dosing schedules and manual triggers, and servo control to dispense exact pill counts with physical button confirmation.
- Defined the database schema and the device-to-server API contract tying hardware to the dashboard.

Mars Rover Competition Drone — Northeastern

- Designed the 1.5 kg dummy-weight payload for the team's 3 kg competition drone, optimizing center-of-mass placement and a connect/disconnect mechanism that reduced payload oscillation during flight and release.
- Team placed **6th** in the Mars Rover Competition.

Ride Engineering Competition — Theme Park Engineering Club

- Designed and analyzed an amusement ride to **ASTM F2291**, the industry standard practice for the design of amusement rides and devices. [Add your specific role / result.]

Dating App (in progress)

private repository

- Building a full-stack dating application on the same architecture as RxManager (Next.js front end, Supabase/PostgreSQL back end).

TECHNICAL SKILLS

CAD: SolidWorks, Fusion 360, AutoCAD

Programming & Analysis: MATLAB, Python, MicroPython

Fabrication: 3D printing, soldering

Tools: Git, SQL / Supabase

LEADERSHIP & ACTIVITIES

- Cheese Club — Board Member
- Theme Park Engineering Club — Member
- Mars Rover Competition Team — Member